

LLM-powered research planning agent

Planning, retrieving and screening health-sciences literature with two cooperating agents and a human verification gate

Intelligent Agents · Unit 11 individual presentation · October 2026

Screening is the bottleneck the agent targets

The agent automates

- Turning a question into search subgoals and queries
- Retrieval from PubMed and Semantic Scholar
- Deduplication and first-pass screening
- Extractive, cited summaries

The researcher decides

- Approve, flag or reject every record
- Optionally fix eligibility criteria
- Revise the plan with feedback
- Only approved records enter the evidence table

Out of scope by design

- Clinical recommendations
- Patient-identifiable data
- Claims from paywalled full text

Success criteria fixed in the proposal

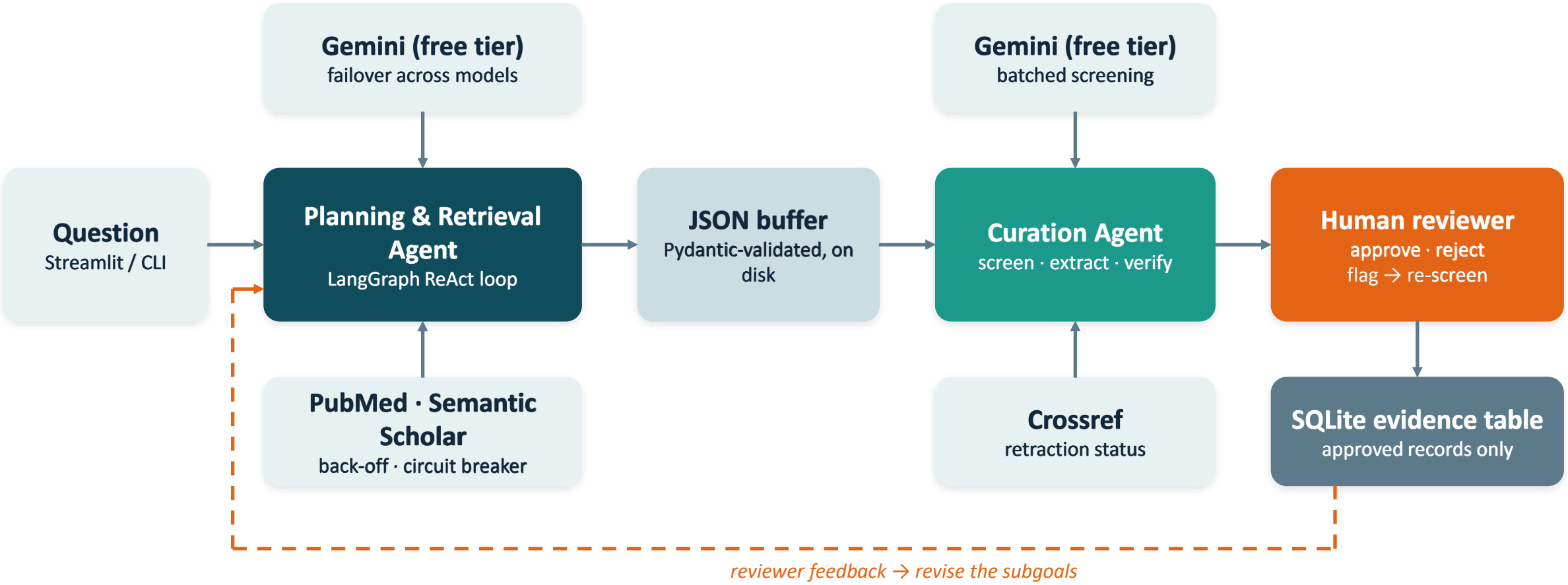
Recall
≥ 0.80 vs a labelled set

Traceability
every claim → ID + extract

Fidelity
no claim absent from abstract

Storage
store reconciles with decisions

Two cooperating agents and a human gate



Hybrid: deliberate once at decomposition, react during retrieval (Wooldridge, 2009). A validated JSON buffer replaces an agent communication language such as KQML: both agents share one trusted process and a fixed schema (Finin et al., 1994).

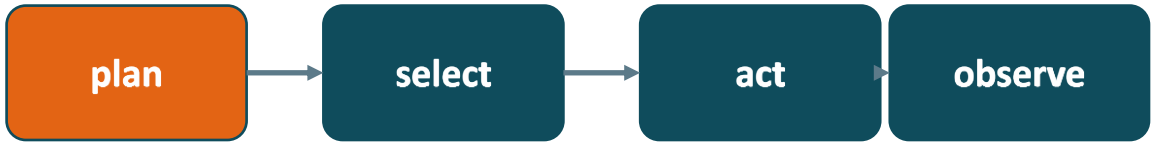
Planning agent: BDI intentions, ReAct execution

BDI as a conceptual model (Rao and Georgeff, 1995)

Beliefs	records retrieved + query log
Desires	the research question
Intentions	committed subgoal queue; commitment bounds deliberation (Bratman, Israel and Pollack, 1988)

Code: AgentState in research_agent/planning.py

ReAct loop as a LangGraph state machine (Yao et al., 2023)



↺ refine → act · next subgoal → select · done → end

LLM consulted only on anomalous observations
Expected results → rule-based thought; too few → LLM chooses refine / accept / drop

Validated plans, bounded retries
Pydantic schema; 3 re-prompts with the error, then escalate to the human

Circuit breaker and step budget
A source failing twice is skipped (Nygard, 2018); unfinished subgoals are reported as failed

Curation Agent: every claim is checked against its source

1 Deduplicate RapidFuzz + IDs

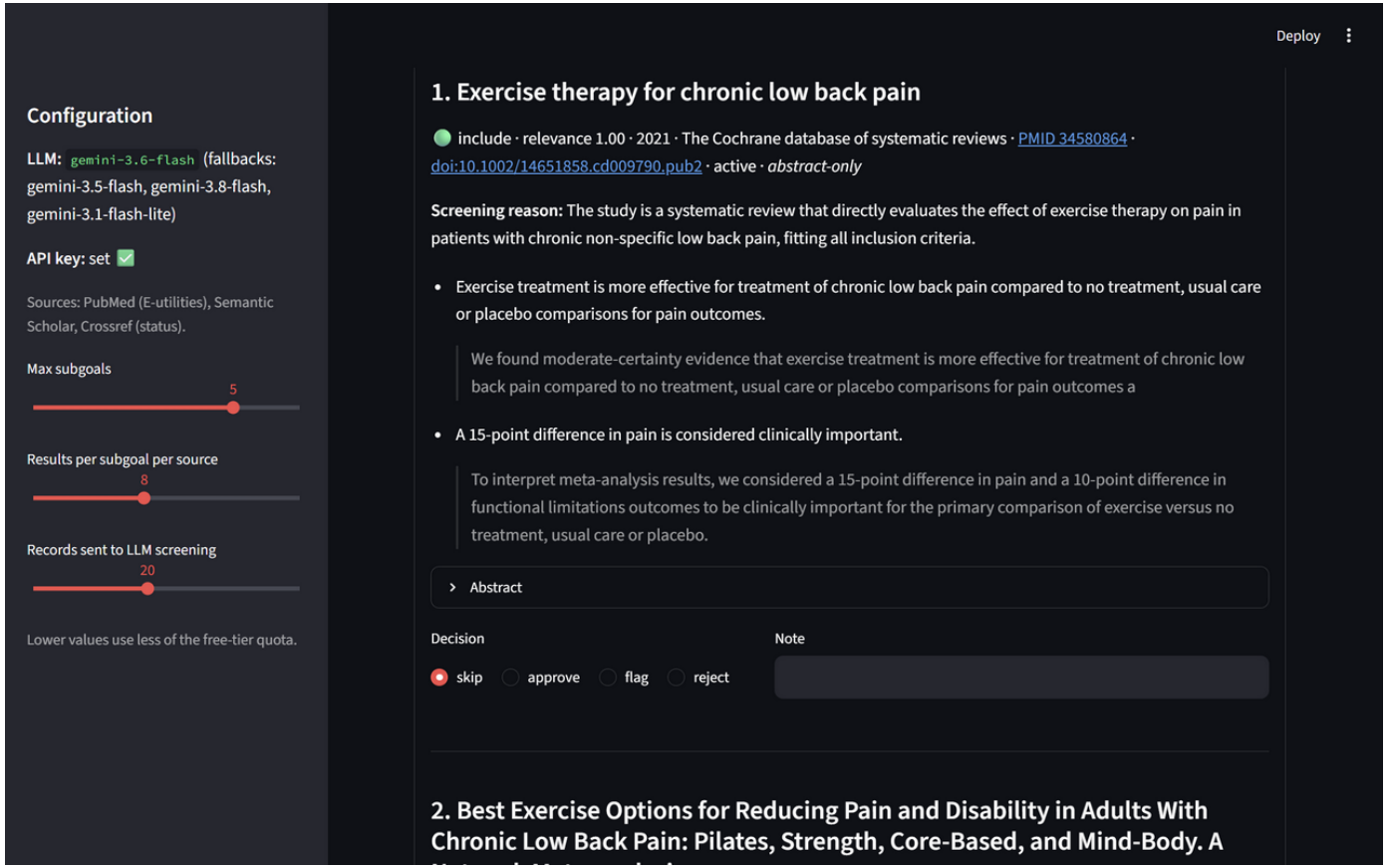
2 Pre-rank BM25

3 Screen LLM, batches of 5

4 Verify extracts word-level, in code

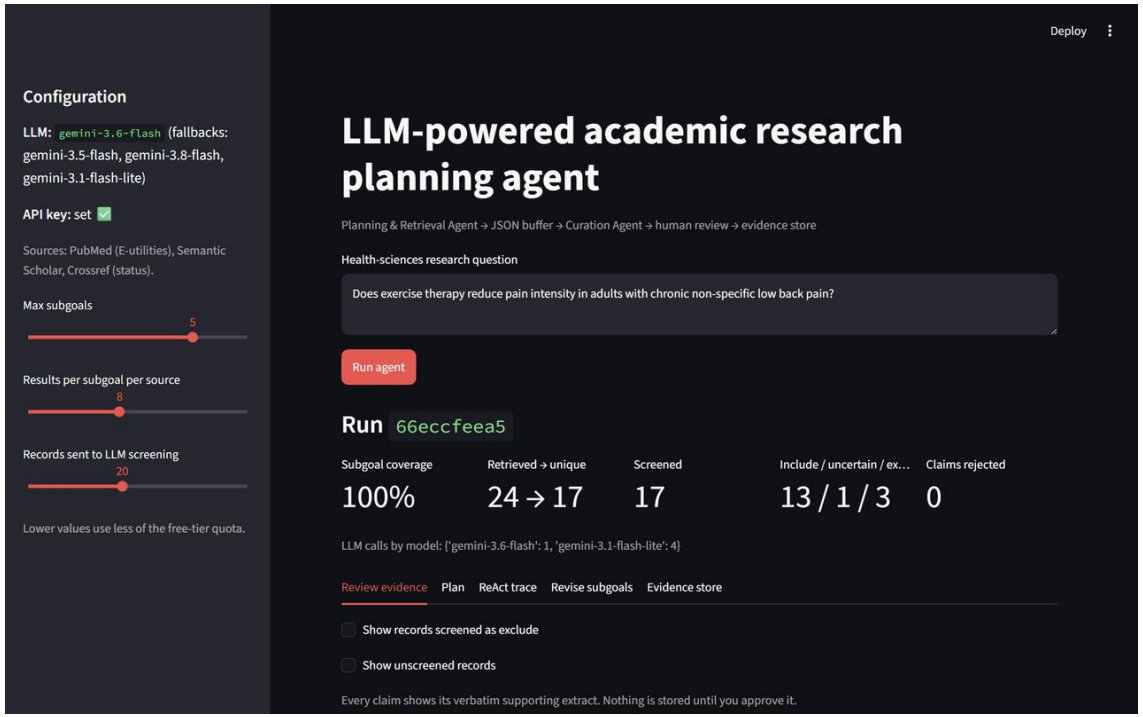
5 Status Crossref

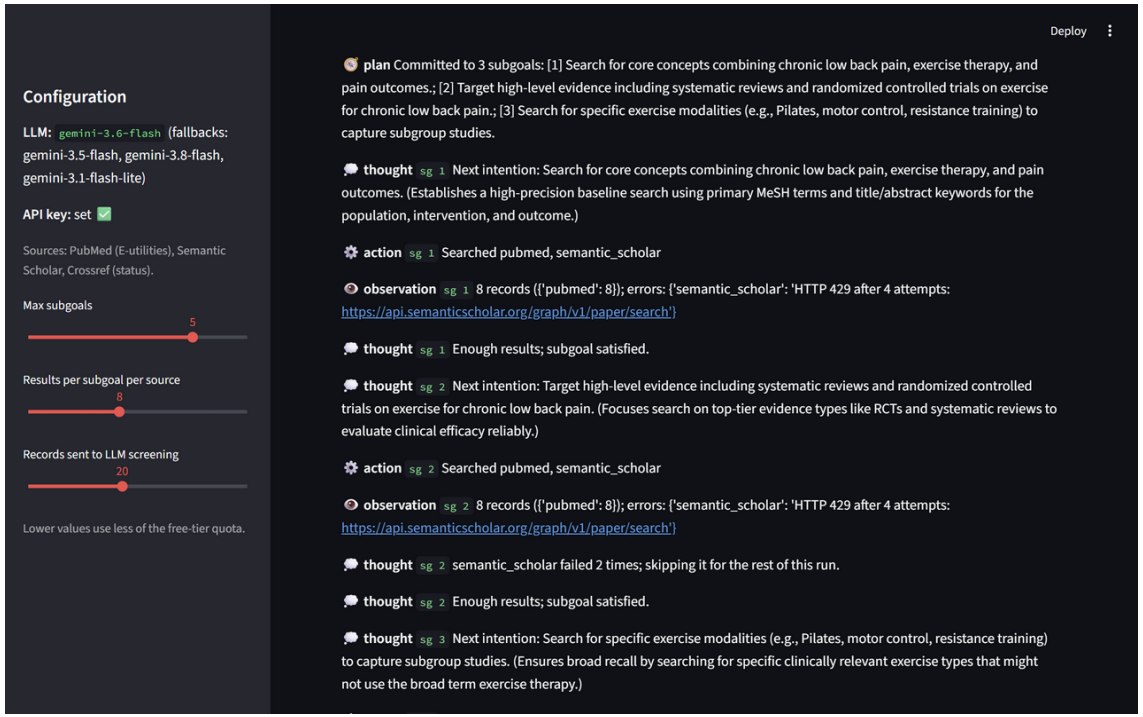
Only step 3 uses the LLM. Numbers, negations and direction words must match exactly; an unverifiable include becomes uncertain.



Live review card: each claim beside its verbatim extract, PMID/DOI links, Crossref status, abstract-only label (Wang, Z. et al., 2025; Wang, G. et al., 2026)

Live run: working around a failing source





100%

subgoal coverage (3/3)

24 → 17

retrieved → unique

13 / 1 / 3

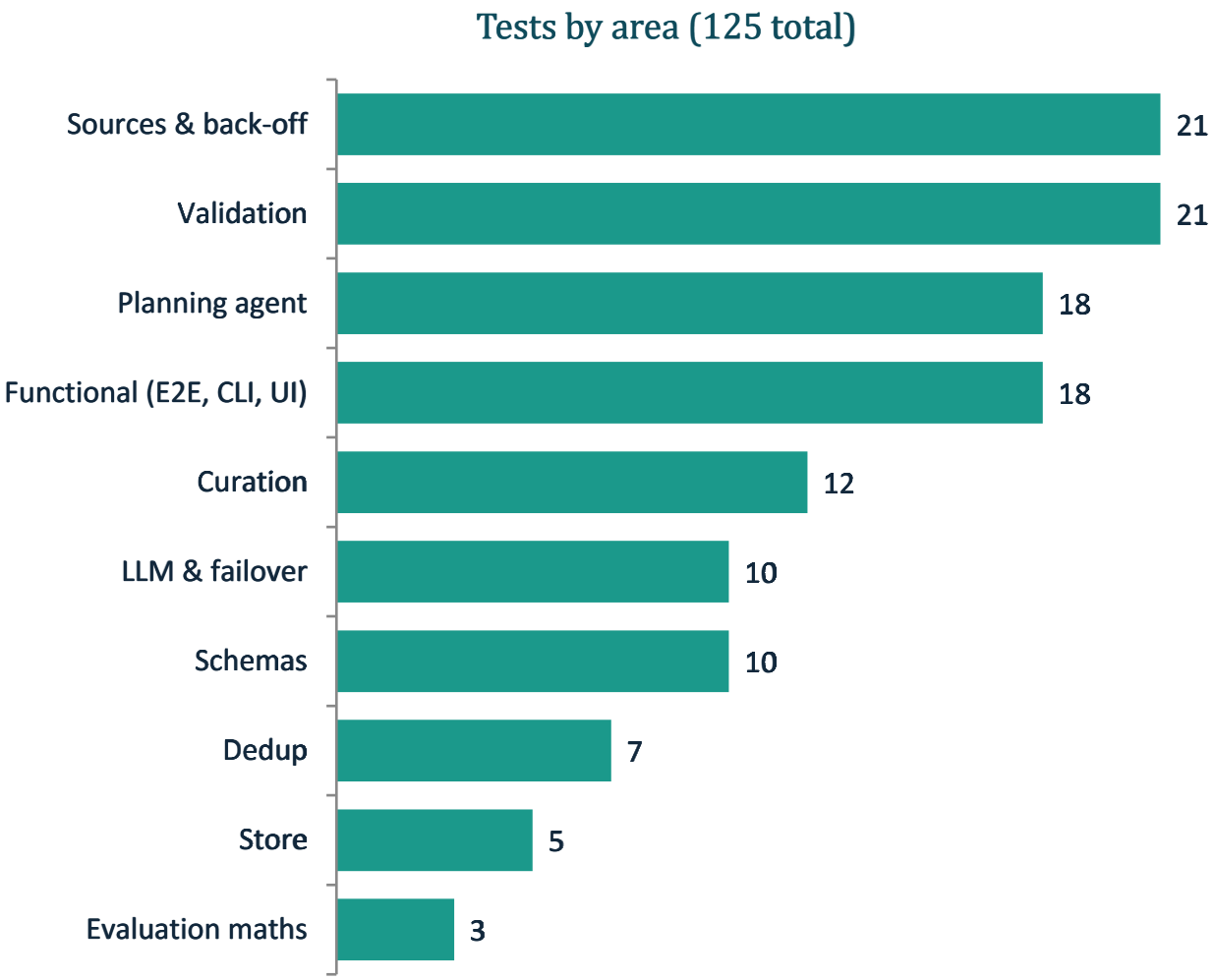
include / uncertain / exclude

3.6 → 3.1 Lite

LLM failover, recorded per call

Right: Semantic Scholar returned HTTP 429 twice and the circuit opened. This run still showed 100% coverage; the review caught it (R11) and reports now name unavailable sources.

Testing: 125 deterministic tests plus live smoke tests



125 / 125

deterministic tests pass, ~10 s, no network or key

97%

line coverage of research_agent

3 / 3

live smoke tests: PubMed, Crossref retraction, Gemini plan

Fakes, not mocks of internals: real source clients, LangGraph loop and SQLite store; only the network (respx: recorded PubMed, synthetic S2) and the LLM (prompt-aware fake) are replaced. The UI is tested with Streamlit AppTest.

Evaluation against the proposal's success criteria

1.0

Recall: records sent to reviewer
target ≥ 0.80 · same in 3/3 runs

0.9

Recall at the default cut-off
one include ranked below the BM25 top 20

0.833

Specificity
errs towards inclusion (pain sensitisation trial)

3 / 3

Store reconciles
evidence table = approvals in decision log; traceability guaranteed by design

Method: retrieval held fixed; 28 real PubMed records stratified with near-misses (acute pain, neck pain, passive-only therapy, protocols, stenosis); screened three times.

Conclusion, limitations and next steps

Mostly achieved

- Two cooperating agents; hybrid BDI + ReAct planner
- Validated hand-off; word-level verified claims
- Human gate with flag → re-screen loop
- Visible, not silent, degradation

Limitations

- Abstracts only; English filter on PubMed
- Semantic Scholar unavailable without a key
- Validator proves extracts, not paraphrases

Next steps

- Human-screened set of 50 records
- Semantic Scholar API key
- Open-access full text (PMC)

Ethics: sanctioned APIs only; abstracts go to a free-tier LLM and stay in run files; only approved records enter the evidence table; defaults counter automation bias (Goddard, Roudsari and Wyatt, 2012).

Code, tests, evidence and README: project repository · Full references in README and transcript